LAB EXPERIMENT RECORD

15-10-2026---Set 20: Stock Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
Name	Dudekula Mohammad Ilyas
Register Number	192325017
Date	15 OCTOBER 2026
Day	THURSDAY

Aim

To analyze the prices of three stocks using a line chart, bar chart and table, and to build an interactive Tableau dashboard for exploring the stock prices.

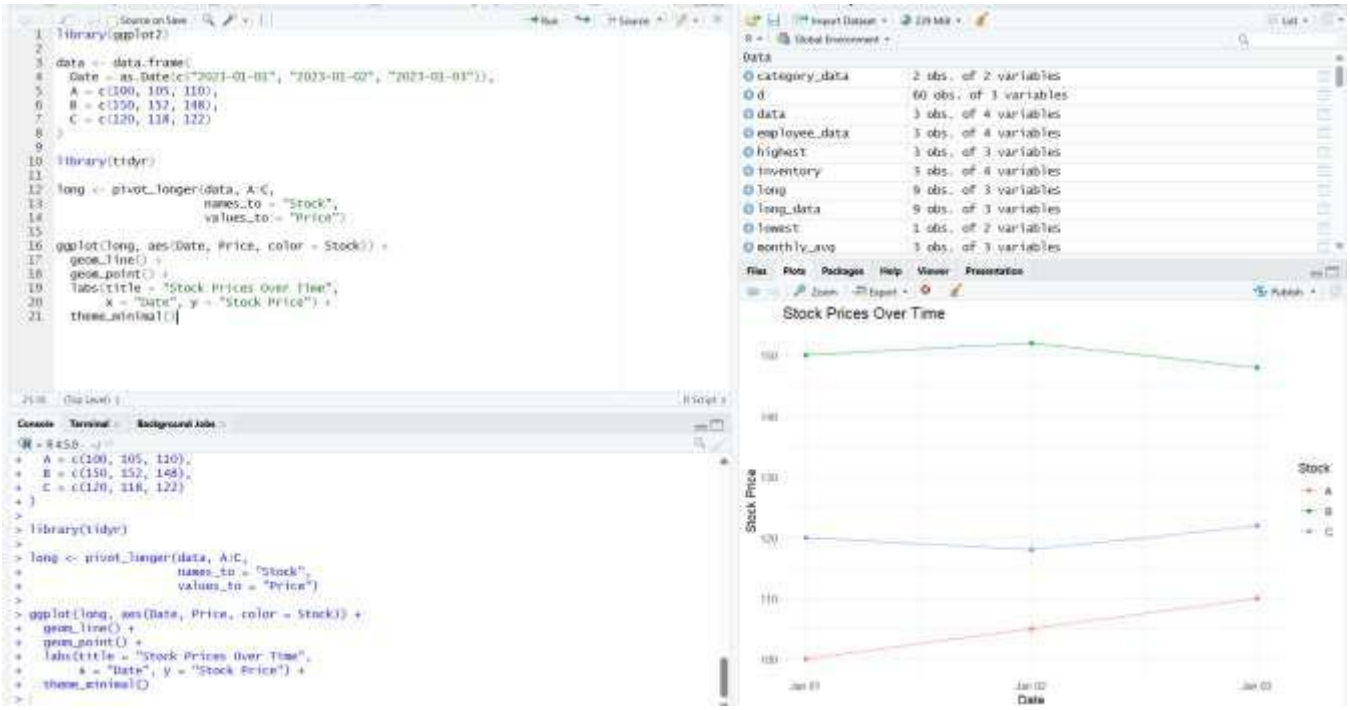
Dataset: Stock Prices

Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

Question 1: Line Chart of Stock Prices

Question: Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.

Aim: To compare the prices of Stock A, Stock B and Stock C over the three days using a line chart with labelled axes and a title.

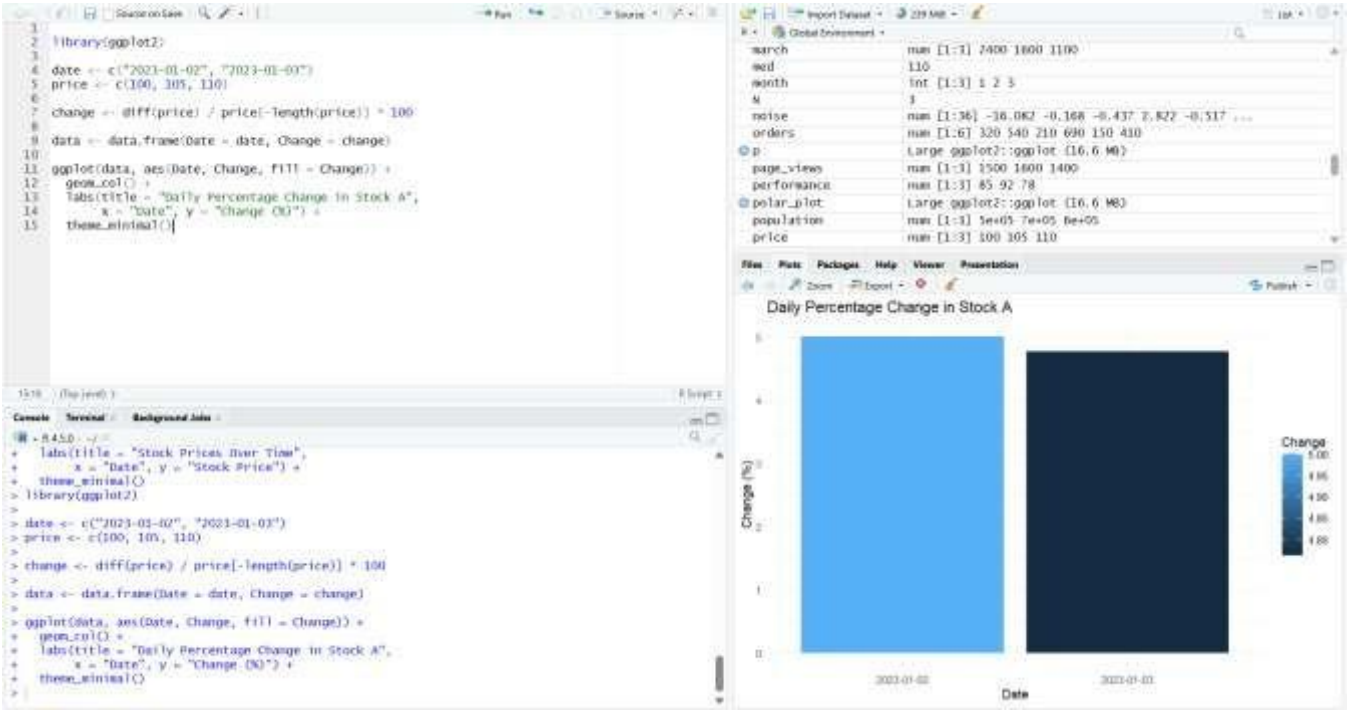


Result: Stock A rises steadily from 100 to 110, Stock B stays the highest (148 to 152) with small changes, and Stock C moves slightly between 118 and 122.

Question 2: Bar Chart of Daily Percentage Change for Stock A

Question: Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.

Aim: To show how much Stock A changed each day, in percent, using a labelled bar chart.

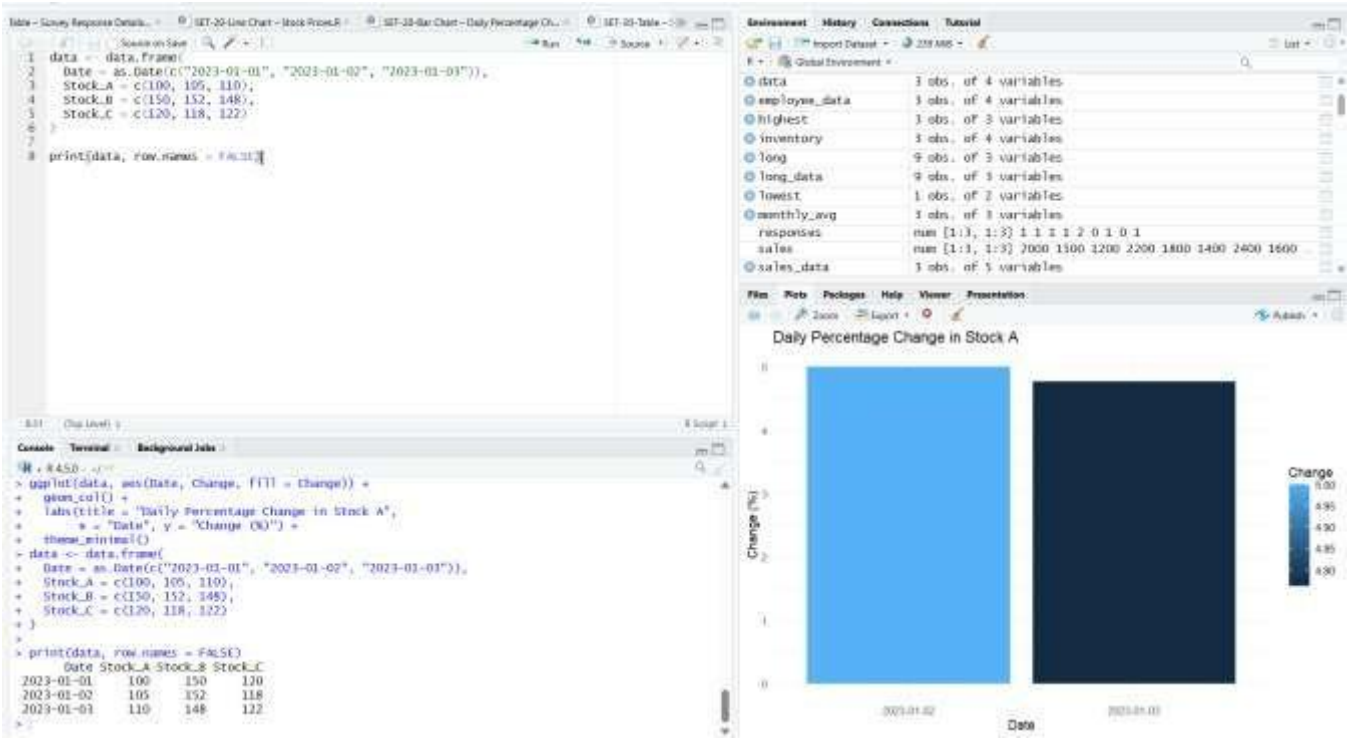


Result: Stock A rose by 5.0% on 2023-01-02 (100 to 105) and by about 4.8% on 2023-01-03 (105 to 110); the first day has no earlier price, so it has no change.

Question 3: Table of Stock Prices

Question: Build a table to display the stock price data for each company over the given period. Label the table elements.

Aim: To present the daily price of each stock in a clearly labelled table.

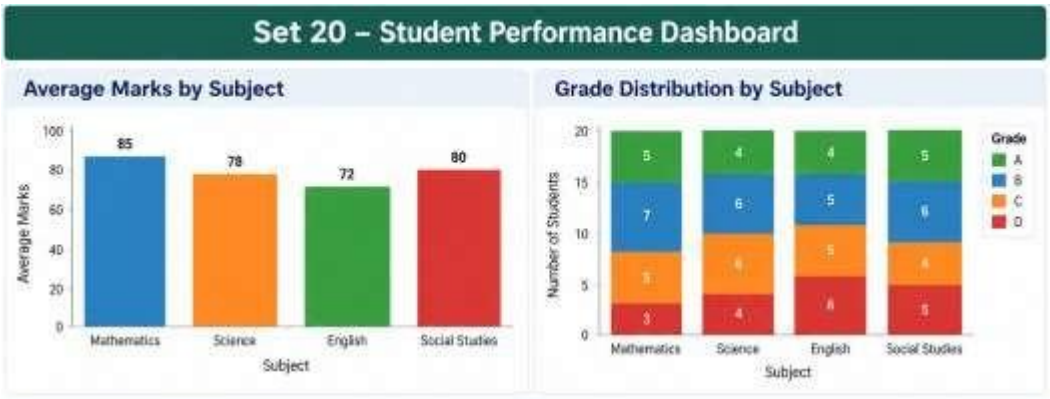


Result: The table lists all three days; Stock B has the highest price on every day, and on 2023-01-03 the prices are 110, 148 and 122.

Question 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.

Aim: To build an interactive Tableau dashboard that combines the line chart, bar chart and table for exploring stock prices.



Student Performance Details

Student ID	Student Name	Mathematics	Science	English	Social Studies	Total Marks	Average Marks	Grade
S001	Aditi	88	76	82	79	325	81.25	A
S002	Bhanu	92	85	78	88	343	85.75	A
S003	Chitra	75	69	70	72	286	71.50	B
S004	Dinesh	68	72	65	70	275	68.75	B
S005	Eswar	95	90	85	92	362	90.50	A
S006	Farhan	80	77	68	75	300	75.00	B
S007	Gowri	60	65	58	62	245	61.25	C
S008	Harsha	78	74	80	76	308	77.00	B

Result: The dashboard places all three views together, and a date filter updates them so users can explore the prices interactively.

Overall Result

Observation: Over the three days Stock A gained 10% (100 to 110), Stock C gained about 1.7% (120 to 122) and Stock B fell about 1.3% (150 to 148).